



SOH prediction of lithium battery based on IC curve feature and BP neural network

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ARTICLE INFO

Keywords:

Lithium battery
State of health
Prediction model
Increment capacity
Neural network

ABSTRACT

Precise battery SOH (state of health) prediction and monitoring are of extreme importance for the future intelligent battery management system (BMS). In this paper, battery discharge experiments at different temperatures were carried out. A battery SOH prediction model based on incremental capacity analysis and BP neural network is proposed to predict battery SOH at different ambient temperatures. By analyzing the correlation between the characteristics of IC curve and SOH, the mapping relationship between temperature and IC curve characteristics is established by using the least square method, and the SOH prediction model at different temperatures is obtained. At the same time, combined with ICA, an online real-time correction prediction model is established, and the characteristic data is continuously updated to ensure the SOH prediction accuracy under different aging states. Finally, the feasibility of the prediction method proposed in this paper is verified by comparing the model test results and experimental results, the average error of the model prediction results is 1.16%. Thus, by establishing the relationship between temperature and IC curve characteristics, the battery SOH at different temperatures can be predicted.

1. Introduction

As one of the power sources of electric vehicles (EV), lithium battery has attracted much attention in the automotive industry. Due to the existence of battery aging mechanism, the available capacity of lithium battery will decrease with the increase of service times, and the service life will also decrease [1], which has a great impact on the endurance mileage of electric vehicles. And according to the requirements of United States Advanced Battery Council (USABC), a lifetime of lithium-ion cells of up to 1000 cycles at an 80% depth-of-discharge (DOD) is required [2]. Therefore, it is necessary to estimate the state of health (SOH) of battery for diagnosis, so as to provide follow-up management system for analysis and regulation of vehicle conditions and extend battery life.

Generally, SOH is defined as the ratio of the current capacity and the initial capacity of the battery. However, battery capacity cannot be measured directly, and can only be estimated by basic parameters such as voltage and current. Moreover, lithium batteries experience capacity decay when charging and discharging as well as over time even when they are inactive [3]. Therefore, researchers put forward various algorithms to improve the estimation accuracy. Many of these methods have

achieved good results, which can be basically divided into two categories: the method based on model and data-driven. Among them, the method based on model is to establish electrochemical model or equivalent circuit model to realize SOH estimation with Kalman filter (KF) or Gaussian filter. Zhao and Qiu [4] combines the two models to establish an improved battery model, considering the value of battery internal resistance under different working conditions, an improved SOH estimation method based on the unscented Kalman filter (UKF) algorithm is proposed. Xu et al. [5] established an equivalent circuit model considering the influence of capacity rebound and internal capacitance resistance, obtained the relationship between internal resistance and SOH through experiments, and predicted the battery state using dual extended Kalman filter (DEKF). The data-driven method base refers to learning the mapping relationship between features and SOH through neural network and support vector machine [6] or constructing soft sensors for prediction [7]. This method has wide applicability, but feature selection is the key to use this method. Commonly used features include basic parameters, such as voltage and temperature or extreme value of battery temperature curve.

In the method of extracting relevant features, incremental capacity curves have better results, which is introduced in literature [8–12]. It is commonly used to obtain the capacity increment curve by differential

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<https://doi.org/10.1016/j.energy.2022.125234>

Received 16 January 2022; Received in revised form 19 August 2022; Accepted 20 August 2022

Available online 26 August 2022

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Nomenclature			
Abbreviation			
BMS	Battery management system	b_{2k}	threshold value of output layer
BP	Back propagation	f_{θ}	Fitting function
CC_CV	Constant current and constant voltage	h	number of hidden layer nodes
DEKF	Dual extended Kalman filter	K	Weight
DOD	Depth-of-discharge	n	Total number of samples
IC	Incremental capacity	p_m	IC curve characteristic
ICA	Incremental capacity analysis	Q_0	Initial battery capacity
KF	Kalman filter	Q_{now}	Current battery capacity
PICC	IC curve peak	r	Pearson correlation coefficient
RUL	Residual service life	x_i	eigenvalue
SOH	State of health	y_i	battery SOH
UKF	unscented Kalman filter	$\bar{x}$	average values of eigenvalue
USABC	United States Advanced Battery Council	$\bar{y}$	average values of battery SOH
Symbols		Greek letters	
a_{1i}	output of the hidden layer	θ_i	polynomial coefficient
a_{2k}	output of the output layer	Subscripts	
b_{1i}	threshold value of hidden layer	i	Number of sample
		k	Number of cell cycle
		m	Number of sample groups

derivation of voltage and capacity curve, and to select more appropriate characteristic values from the curve. The accuracy of feature extraction will directly affect the accuracy of estimation. Some literatures specifically study how to accurately extract effective IC curve features. It was mentioned in literature [13] that Pearson correlation coefficient was used to show the strong correlation between power characteristics and SOH, and the residual life of lithium batteries was predicted by combining neural Gaussian process. Li et al. [14] combines entropy weight method and grey correlation degree to evaluate battery health. In Ref. [15], the IC curve is obtained by numerical derivation, and in order to eliminate the curve noise, a smoothing method based on Gaussian filter is proposed to accurately identify the battery aging characteristics. There are also methods through combination of ICA and data-driven. Such as Weng [16] and others have established two neural network models for estimating SOH and residual service life (RUL) by analyzing aging cycle data and combining IC curve with artificial neural network. The results have high accuracy.

In the study of battery SOH, the temperature will also have an impact on the battery SOH [17]. Tran et al. [18] studied the influence of temperature on equivalent circuit parameters and finds that the three parameters are closely related to temperature. Many literatures use experimental methods to calibrate the SOH at different temperatures and then predict them. However, the changes in the actual work of the battery lead to inaccurate prediction results. ICA is an effective method to estimate battery SOH at different temperatures. Because of the strong correlation between curve characteristics and SOH, the relationship between temperature and capacity can be established indirectly through characteristics. And, IC curve characteristics partially exist on the voltage curve during charging and discharging, it is easy to obtain in practical application. These two points can be used to realize on-line update to ensure the estimation accuracy.

For the SOH prediction of lithium battery at different temperatures, a model based on ambient temperature and ICA is proposed in this paper. The model consists of two parts: one is to fit the characteristics of ambient temperature and IC curve by using least square method; and the other is to predict battery SOH by combining BP neural network. There are three specific contributions: firstly, the discharge curves of the battery at different temperatures are tested and recorded, and the influence of temperature on the battery capacity is analyzed by comparing the changes of the curves. Then, combined with Pearson index, the feature with the highest correlation is extracted from the obtained IC curve as a

health factor. Secondly, the least square method is used to fit the relationship curve between temperature and health factors. This step is to determine the characteristics of IC curve through temperature, add a correction process to the model to realize online update, and then take the health factors as the input data set of BP neural network learning. Finally, the model test and experimental results are compared to verify the feasibility and accuracy of the method.

The rest of this paper is organized as follows. Section 2 introduces the battery experiment and extracts the characteristics of IC curve. Section 3 establishes the relationship between IC curve characteristics and temperature by least square method, and constructs SOH prediction model combined with BP neural network. In Section 4, the model test results and experimental results are compared to verify the feasibility of the model. The main conclusions are given in Section 5.

2. Discharge experiment and feature extraction of lithium battery at different temperatures

2.1. Battery discharge experiment

To analyze the effect of temperature on battery discharge, it is necessary to measure the battery capacity and charge-discharge curves at different temperatures. In this paper, five 18650 ternary lithium batteries are selected, labeled 1, 2, 3, 4, and 5, and are connected in series on the experimental equipment for charge and discharge experiments. The batteries were five lithium-nickel-manganese-cobalt (LiNMC) batteries and their parameters are shown in Table 1. The test bench that was used to acquire the required experimental data is shown in Fig. 1, including charge and discharge tester, temperature control box and a control computer. During the whole experiment, the battery was exposed to a constant temperature and humidity chamber, the charging

Table 1
Battery parameters used in the experiment.

Project	Parameter
Nominal capacity	3000mAh
Cell nominal voltage	3.6V
Charging cut-off voltage	4.2V
Discharging cut-off voltage	2.75V
Size	65.2mm × 18.5 mm
Charging temperature	0 °C–40 °C

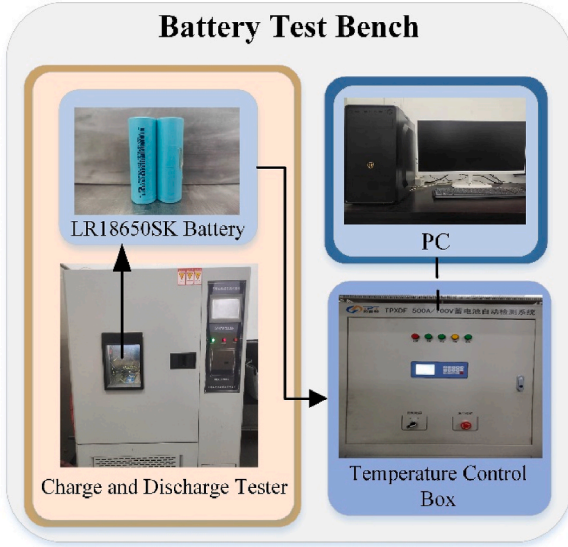


Fig. 1. Battery test bench.

method is constant current and constant voltage, and the discharging method is constant current, the data sampling interval was 1s.

In order to prevent the actual capacity of the battery from being inconsistent with the nominal capacity, the capacity of the battery shall be tested before the beginning of the experiment. The capacity presented by the battery is related to the discharge rate. C rate is used to describe the size of the discharge rate, which is the ratio of the current to the nominal capacity. The standard 1C discharge rate is used to measure the battery capacity. The detailed process is as follows: 1. Firstly, to avoid the influence of excessive discharge rate on battery capacity, discharge at a rate of 0.2C to the cutoff voltage of 2.7V and stand for more than 1h. 2. Use the constant current and constant voltage (CC-CV) charging mode to charge the battery at 0.5C rate, the charge cut-off voltage is 4.2V and the cut-off current is 0.5A. 3. Discharge with constant rate 1C to the discharge cutoff voltage, test result display the measured capacity basically conforms to the nominal capacity. The whole test process was carried out at 25 °C.

In this paper, the temperature control box is used for constant current discharge test at different temperatures. After the temperature was set, the battery was left standing for more than 30min, and then charged and discharged with rate 1C according to the above method, and recorded the data. The discharge data onto 8 temperatures are selected to analyze the change of battery capacity. Fig. 2 shows the voltage-capacity curve of No. 1 battery at different temperatures. It can be seen that the battery discharge capacity increases with the increase in temperature, because the charge and discharge process of lithium battery is a chemical reaction process of lithium ions in Extraction and insertion, and the increase in temperature will accelerate the gain and loss rate of electrons between the positive and negative, the polarization of the battery decreases and the internal resistance decreases [19]. It is worth noting that the higher the temperature in the voltage range from 3V to 3.6V, the curve shows a downward trend and the inflection point position increases, which lays a foundation for the subsequent study of the relationship between temperature and battery SOH through ICA.

The formula for calculating SOH is given as follows :

$$SOH = \frac{Q_{now}}{Q_0} \times 100\% \quad (1)$$

Where Q_{now} represents current battery capacity, Q_0 is initial capacity.

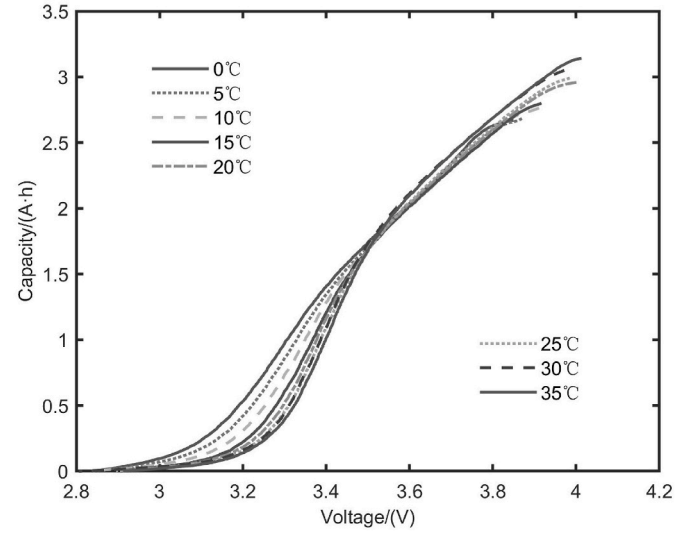


Fig. 2. Voltage-capacity curves.

2.2. IC curve acquisition and feature extraction.

The concept of incremental capacity analysis was first proposed by Thompson in 1979 and applied to material research [20], and it has only been used in battery SOH estimation in recent years. In the study of lithium battery capacity degradation, IC curve represents the increment of battery power in unit voltage. Theoretically, it is derived from U-Q curve, but in practical application, the derivation of discrete data is usually calculated by difference method, and the formula is shown in Eq. (2):

$$IC = \frac{dQ}{dU} = \frac{\Delta Q}{\Delta U} = I \cdot \frac{\Delta t}{\Delta U} \quad (2)$$

Where Q is the battery capacity, U is the voltage and I is the battery current.

Due to the influence of the sensor accuracy of the experimental equipment and the instability of the internal chemical reaction of the battery, the measured data often contain noise, which will be amplified by differential calculation, making it difficult to obtain effective features of the IC curve. To this end, this paper uses the LOESS smooth function in MATLAB for fitting. Fig. 3 shows the IC curve of No. 1 battery at different

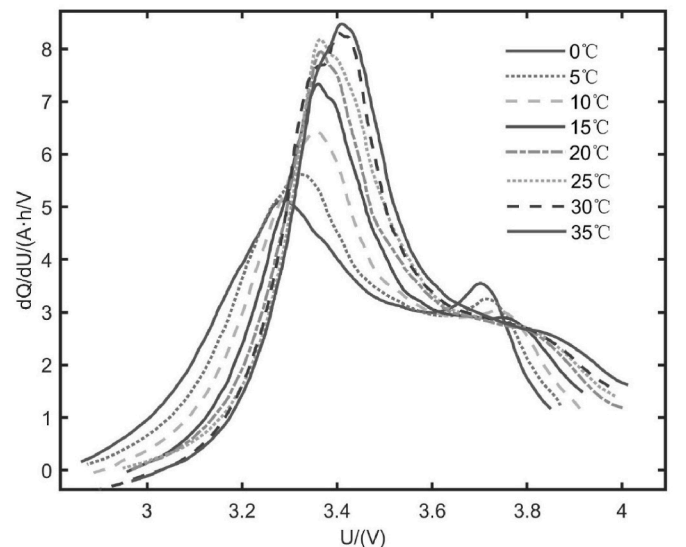


Fig. 3. IC curves.

temperatures.

It can be seen from the figure that with the increase in ambient temperature, the whole curve shifts to the right, the peak of the curve becomes larger, and the position of the peak appears at a higher voltage. These changes can be mapped to the battery capacity. In order to obtain this mapping relationship, it is necessary to obtain a characteristic value of the curve that can be closely related to the battery SOH as a health factor. Generally, the features that can be extracted from the IC curve include peak (PICC), peak position and peak slope [21], etc. Different features have different degree of correlation with SOH, and the higher the correlation degree of features, the greater the impact on SOH [22]. In order to improve the estimation accuracy, this dissertation put into use of Pearson Correlation analysis to verify the correlation between the above two. Pearson correlation coefficient is calculated using Eq. (3):

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad (3)$$

Where r is the correlation coefficient and n is the number of samples; x_i is the eigenvalue, y_i is battery SOH; $\bar{x}$, $\bar{y}$ they are the corresponding average values. The judgment method is that the greater the value of $|r|$ is, the higher the degree of correlation is considered. The correlation coefficient between characteristics and SOH is calculated, and the results are shown in Table 2.

From the data in Table 2, the peak value has the highest correlation with SOH. In addition, literature [23] mentioned that the peak in IC curves is related to the internal active material of lithium battery, which can accurately reflect the battery aging. Therefore, PICC is adopted as the health factor in this paper.

3. SOH prediction strategy at different temperatures

3.1. Modeling of mapping relationship between temperature and IC curve characteristics

Temperature is an important factor affecting the discharge performance of the battery. When the battery is at different ambient temperatures, the energy released by the battery is different. Estimating SOH at current temperature by ICA method can directly measure and calculate IC curve, but it cannot predict SOH at other temperatures. This defect can be made up by connecting the characteristics of temperature and IC curve. In this paper, SOH estimation of different temperatures is achieved by fitting the characteristics of different temperatures. The characteristics and temperatures in the five batteries in the experiment are shown in Fig. 4.

It can be seen from the figure that the characteristic values of five batteries at the same temperature are different, which is caused by the differences between batteries and the errors in the process of measurement and calculation. Reducing these errors will greatly improve the fitting accuracy. By comparing the discharge curves of five batteries at 25 °C, it can be seen that the difference between different batteries is very small, and the error can be ignored. Excluding battery differences, these data can be fitted with the least square method to obtain a curve to represent the relationship between temperature and characteristics.

Based on the least square fitting principle [24], the objective function is established:

Table 2
Pearson Correlation coefficient of characteristics.

Battery	Peak position	Peak left slope	PICC
1	0.875	0.901	0.926
2	0.831	0.866	0.915
3	0.799	0.889	0.896
4	0.823	0.913	0.908
5	0.851	0.898	0.903

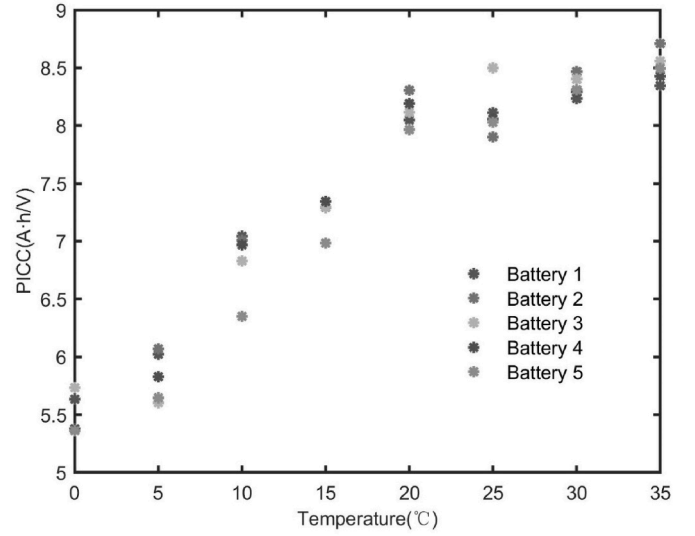


Fig. 4. Temperature and IC curve characteristics.

$$f_{\theta}(x) = \theta_0 + \theta_1 x + \theta_2 x^2 + \dots + \theta_n x^n \quad (4)$$

Where θ_i is the polynomial coefficient.

By finding a set of $\theta(\theta_0, \theta_1, \theta_2, \dots, \theta_n)$, the sum of squares of error is minimized.

$$\sum_{i=1}^n r_i^2 = \sum_{i=1}^n (f_{\theta}(x_i) - y_i)^2 \quad (5)$$

Where n is the polynomial order and y_i are the actual values of the sample.

Since one temperature corresponds to multiple eigenvalues in this paper, formula (2) is modified as follows:

$$\sum_{i=1}^n r_i^2 = \sum_{i=1}^n [(f_{\theta}(x_i) - p_1(x_i))^2 + (f_{\theta}(x_i) - p_2(x_i))^2 + \dots + (f_{\theta}(x_i) - p_m(x_i))^2] \quad (6)$$

Where m is the number of sample groups, x_i and $p_m(x_i)$ respectively temperature and selected feature. The experimental data are substituted into Formula (3) to obtain the characteristic curve between temperature

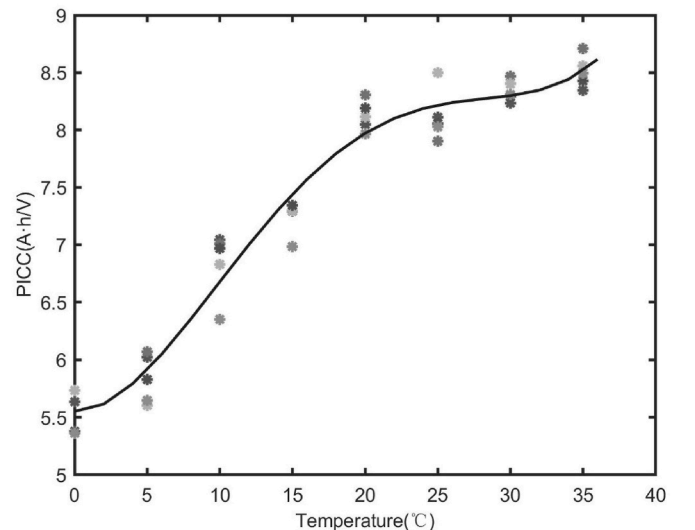


Fig. 5. Fitting results.

and IC, as shown in Fig. 5.

Increasing the amount of data can improve the fitting accuracy. Record the available characteristic data each time during the use of the battery, and add it to the input of the fitting algorithm to improve the accuracy. However, with the increase of battery charge and discharge cycles, the loss of active materials of lithium battery increases [25], the sensitivity to temperature decreases, and the temperature characteristic curve changes, which needs to be adjusted. Therefore, it is necessary to update the data and discard the battery history data in time, so as to accurately predict the battery capacity for different temperatures in the current environment, so as to improve the adaptability of this method. However, in the process of battery SOH prediction, the recent historical data is still valuable in correcting the fitting error, so it should not only need to avoid the saturation phenomenon caused by too much historical data, but also ensure the fitting accuracy. Therefore, this paper uses the method of adding weights to the fitting data to realize the use of historical data and avoid the saturation phenomenon.

First, take the data of the offline fitting algorithm as the initial input data, as shown in Formula 3.

$$\begin{bmatrix} p_1(x_1) & p_1(x_2) & \cdots & p_1(x_n) \\ p_2(x_1) & p_2(x_2) & \cdots & p_2(x_n) \\ \vdots & \vdots & \ddots & \vdots \\ p_m(x_1) & p_m(x_2) & \cdots & p_m(x_n) \\ x_1 & x_2 & \cdots & x_n \end{bmatrix}$$

Because only one set of data is updated in each cycle, and the parameter temperature in the updated data is determined by the battery environment and is random, the data updated in each cycle is more one-dimensional than the last data. For example, the temperature x_{n+k} and corresponding feature $p_{m+k}(x_{n+k})$ is updated in the k -th cycle. Where $p_{m+k}(x_1), p_{m+k}(x_2), \dots, p_{m+k}(x_{n+k-1})$ loops by $k-1$ fitting results to calculate $p_{m+k-1}(x_i), p_1(x_1), p_{m+k}(x_2), \dots, p_{m+k}(x_n)$ is obtained by calculating $p_i(x_{n+k})$ from $k-1$ cycle fitting results, and the following matrix is obtained:

$$\begin{bmatrix} p_1(x_1) & p_1(x_2) & \cdots & p_1(x_{n+k}) \\ p_2(x_1) & p_2(x_2) & \cdots & p_2(x_{n+k}) \\ \vdots & \vdots & \ddots & \vdots \\ p_{m+k}(x_1) & p_{m+k}(x_2) & \cdots & p_{m+k}(x_{n+k}) \\ x_1 & x_2 & \cdots & x_{n+k} \end{bmatrix}$$

In this way, each update not only retains the original data, but also records the number of cycles k and the updated data, and then takes it as the next input for continuous iteration.

Add weight in formula (3), which can be expressed as:

$$\sum_{i=1}^n r_i^2 = \sum_{i=1}^{n+k} [K(1)(f_{\theta}(x_i) - p_1(x_i))^2 + K(2)(f_{\theta}(x_i) - p_2(x_i))^2 + \dots + K(+k)(f_{\theta}(x_i) - p_{m+k}(x_i))^2] \quad (7)$$

In formula (6), through the influence of the added weight K , the proportion of the error between the actual value and the fitting value in formula (3) is different under different cycle times. The larger the cycle times k is, the higher the proportion will be, so as to enhance the role of new data in fitting.

3.2. SOH prediction modeling based on BP neural network

BP neural network is a multi-layer feedforward neural network, it includes input layer, hidden layer and output layer. The neurons between adjacent layer are connected, the connection strength constitutes a weight w of the network [26]. The simple three-layer network structure is shown in Fig. 6.

The output of the input layer can be expressed as:

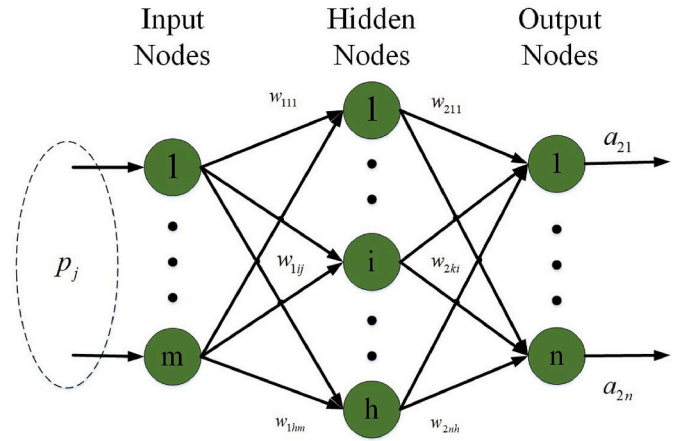


Fig. 6. Structure of BP neural network.

$$a_{1i} = f_1 \left(\sum_{j=1}^m w_{1ij} p_j + b_{1i} \right), i = 1, 2, \dots, s1 \quad (8)$$

The output of the output layer can be expressed as:

$$a_{2k} = f_2 \left(\sum_{i=1}^h w_{2ki} a_{1i} + b_{2k} \right), k = 1, 2, \dots, s2 \quad (9)$$

Where p_j is the input of the j th neuron in the input layer, a_{1i} is the output of hidden layer containing node i th, a_{2k} is the output of the output layer node k , w_{1ij} is the connection weight of node i and node j , w_{2kj} is the connection weight of node i and node k , b_{1i} is the threshold value of hidden layer node i , b_{2k} is the threshold value of output layer node k , f_1 , f_2 and are the transfer functions of the hidden layer and the output layer respectively.

The calculation formula for the number of nodes in the hidden layer is:

$$h = \sqrt{m + n} + a \quad (10)$$

Where h is the number of hidden layer nodes, m and n are the number of input layer and output layer nodes respectively, and a is the adjustment constant of hidden layer, usually ranging from 1 to 10. Since there is only one input and output of BP neural network, h is calculated according to formula (10), and the value range is 2–11. After testing, when $h = 2$, the error of model prediction result is the smallest.

The error calculation formula is:

$$e = \frac{1}{2} \sum_{i=1}^n (y_i - y_i^*)^2 \quad (11)$$

The neural network error continues the process of information forwards propagation and error back propagation within the set learning times, and stops when the error reaches the specified range.

To achieve battery SOH prediction, the PICC is used as the input of the BP neural network, and the output is the battery SOH for learning. There are two functions in the BP neural network that need to be defined by the user, tansig is used as the network activation function, and the output layer uses purelin as the transfer function. Since there is only one input and output of this neural network, according to formula (10), it is calculated that the value of h ranges from 2 to 11. After testing, when $h = 5$, the error of the model prediction result is the smallest. The neural network is trained by gradient descent method, the number of training is 1000 times and the learning rate is set to 0.01.

4. Model validation

The whole prediction strategy includes the relationship model

between temperature and IC curve characteristics and the SOH prediction model based on BP neural network. The BP neural network models require input data in advance for learning, including features and battery SOH. The data of the five batteries mentioned in the experiment at 8 temperatures are used as learning samples. Fig. 7 shows the training state and best validation performance when the experimental data of No. 1 battery is used as the target output. The error histogram and the regression relationship between the actual and target output is shown in Fig. 8 (a) and (b), respectively. The coefficient of determination of this network is 0.8885. The above results prove that the neural network has sufficient learning accuracy.

In order to verify the feasibility of the strategy, within the allowable working range of the battery, the temperature is used as the input of the strategy, and the error between the model output results and the experimental results is calculated and compared. The experiments in the verification process are divided into two parts, one is the charge-discharge experiment of the battery without aging cycle at different temperatures. The other is the charge-discharge experiment at different temperatures of the battery after multiple charge-discharge cycles. The

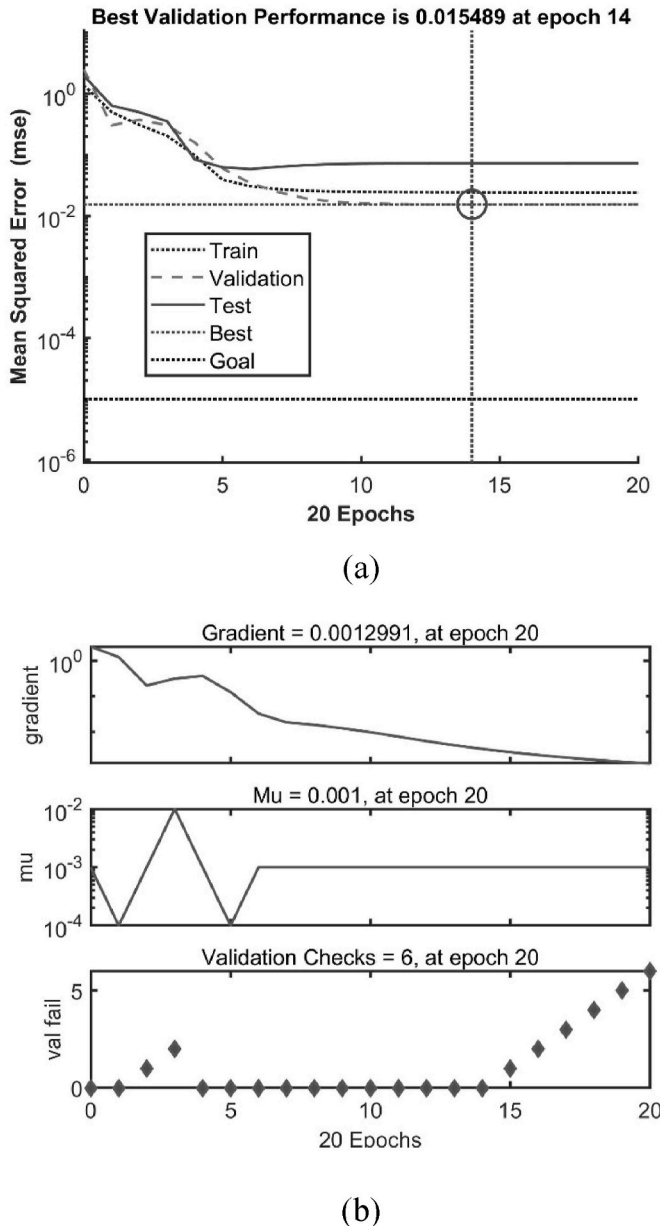


Fig. 7. (a) Best validation performance (b) Training state.

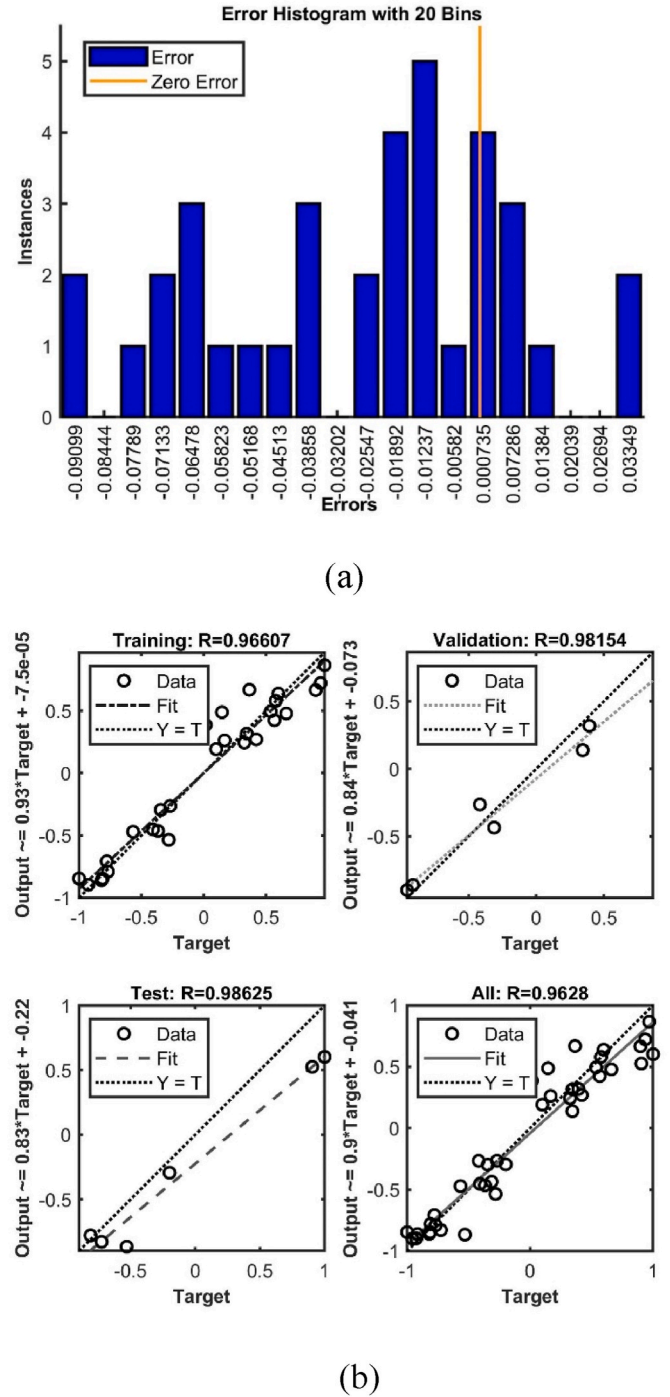


Fig. 8. (a) Error histogram (b) Regression.

former proves that it is feasible to establish the relationship between temperature and SOH indirectly through IC curve characteristics. The latter proves that the method is still valid after a large change in battery capacity.

In the experiment, batteries 1 to 4 were used to test the data without aging cycles, and the number of charge and discharge cycles of these four batteries is less than 45. Fig. 9 shows the predicted results and experimental results of the four batteries in the experiment at 36 temperatures. Through comparison, it can be seen that the predicted results are basically consistent with the trend of the real value of the experiment. The predicted value fluctuates up and down in the real value without too much deviation. It should be noted that when the

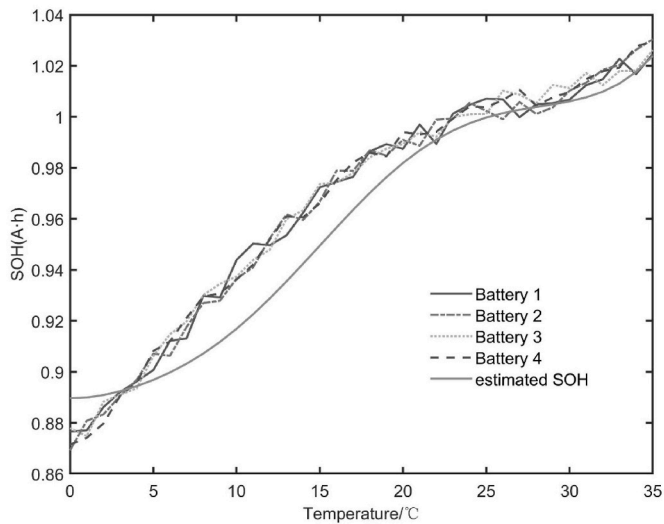


Fig. 9. Model and experimental results of 1–4 battery.

temperature is above 25 °C, it is obvious that there is a large deviation from the previous battery test results. It can be seen from the comparison between the peak value of the curve in Fig. 2 that this is caused by the deterioration of the stability of the battery at high temperature. Fig. 10 shows the error of the experimental and model prediction results of No. 1–4 batteries, in which the maximum absolute error is no more than 3.1%, and the average error of the four batteries is 1.16%. According to the above results, when the battery capacity does not change greatly, the method proposed in this paper can realize the prediction of battery SOH at different temperatures.

Fig. 11 shows the model test results and experimental results of No. 5 battery after 90 charge and discharge cycles at different temperature. The battery has different degrees of over discharge process. At this time, the No.5 battery capacity at 25 °C is 96.45% of the previous one, with a certain loss. The prediction results in the figure have changed significantly compared with the previous ones, which shows that the model records the previous temperature and capacity data and updates the fitting curve, indicating the effectiveness of the online update of the model. The maximum error of the model prediction result is 3.05%, and the average error is 1.26%. This method is still effective after the battery capacity is reduced. It can be seen in the figure that the maximum error

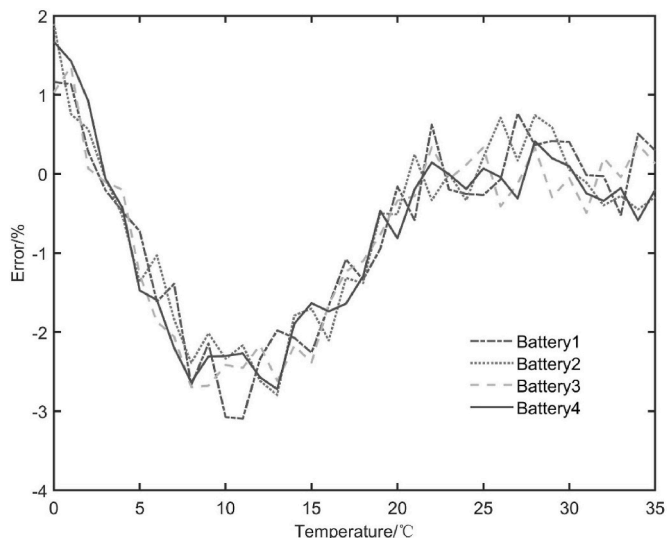


Fig. 10. Battery prediction error.

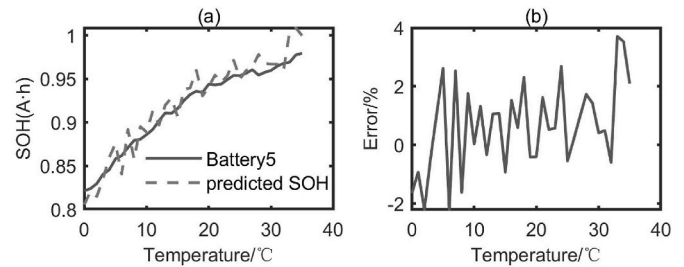


Fig. 11. Prediction results and error diagram of No. 5 battery model. (a) prediction results. (b) error diagram.

Table 3

Prediction error.

Battery	Max error/%	MAE/%	MSE/%
1	3.09	1.18	0.22
2	2.71	1.13	0.20
3	2.69	1.16	0.22
4	2.72	1.16	0.22
5	3.05	1.13	0.21

occurs at 35 °C, which is due to the fact that the battery has not updated the data above 30 °C when the battery is charged and discharged at a room temperature of no more than 30 °C. This also confirms the importance of updating the relationship between temperature and capacity.

To more clearly demonstrate the battery validation effect of this method, Table 3 shows the maximum prediction error, mean absolute error MAE and mean square error MSE. For convenience of presentation, MSE is used to describe the prediction accuracy of BP neural network when capacity is the output. These data show that the BP neural network model has good prediction results.

5. Conclusion

In this study, lithium battery characteristics experiments were conducted at different temperatures, and it was found that the capacity exhibited by the battery had a great relationship with the temperature. After the battery experienced a certain capacity loss, the relationship between temperature and capacity also changed. A strategy is proposed to update the temperature and characteristic curves by recording and fitting IC curve characteristics, and combining with BP neural network to complete the prediction of battery SOH at different temperatures. The strategy is verified by battery experimental data, and the results show that the strategy is feasible. Since this strategy records the IC curve characteristics of the battery during each charge and discharge process, it can better reflect the changes of the battery after multiple aging cycles. In addition, IC curve characteristics are easier to obtain in practical applications than capacity, so it is of practical value to realize the estimation of battery SOH at different temperatures by constructing the relationship between temperature and IC curve characteristics. And the proposed strategy will be helpful for driving planning of electric vehicles in changing environments.

Credit author statement

Xing Chen: Investigation, Writing – Original Draft, Writing – Review & Editing **Jianping Wen:** Conceptualization, Methodology, Formal analysis, Writing – Review & Editing **Xianghe Li:** Visualization, **Yikun Li:** Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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